

BYOUNGWOO PARK

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RESEARCH INTERESTS

Generative AI: Diffusion models, long-horizon video generation, controllable/compositional generation, and generative modeling in function spaces.

AI for Science: Probabilistic modeling and simulation of complex dynamical systems across scales, from molecular and biological processes to weather and climate.

Foundations: Schrödinger bridges, dynamic optimal transport, stochastic optimal control, and sequential Monte Carlo.

RESEARCH EXPERIENCE

New York University

Visiting Scholar

New York, NY

08/2026 - 11/2026

- Probabilistic weather forecasting with distributional diffusion (under review).

Georgia Institute of Technology

Visiting Scholar (Advisor: Prof. [Yongxin Chen](#))

Atlanta, GA

10/2025 - 07/2026

- Dynamic long-horizon autoregressive video diffusion (under review).
- Inference-time sampler for long-horizon compositional diffusion [NeurIPS 2026].

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. in Artificial Intelligence (Advisor: Prof. [Juho Lee](#))

Seoul, South Korea

03/2026 – 08/2028 (expected)

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Artificial Intelligence (Advisor: Prof. [Juho Lee](#))

Seoul, South Korea

03/2024 – 02/2026

- Awards: ICLR 2025 [Oral · Top 1.8%](#)

Inha University

B.S. in Statistics.

Incheon, South Korea

03/2014 – 08/2021

PUBLICATIONS

(* equal contribution, † equal advising)

Conference Papers

[C6] Coarse-to-Fine Compositional Diffusion for Long-Horizon Planning.

Byoungwoo Park, Utkarsh A. Mishra, Jaemoo Choi, Juho Lee, Yongxin Chen
Advances in Neural Information Processing Systems (**NeurIPS**), 2026

[C5] Functional Adjoint Sampler: Scalable Sampling on Infinite Dimensional Spaces.

Byoungwoo Park, Juho Lee, Guan-Horng Liu
International Conference on Machine Learning (**ICML**), 2026

[C4] Stochastic Optimal Control for Continuous-Time fMRI Representation Learning.

Joonhyeong Park*, **Byoungwoo Park***, Chang-Bae Bang, Jungwon Choi, Hyungjin Chung, Byung-Hoon Kim†, Juho Lee†
International Conference on Learning Representations (**ICLR**), 2026.

[C3] Amortized Control of Continuous State Space Feynman-Kac Model for Irregular Time Series.

Byoungwoo Park, Hyungi Lee, Juho Lee
International Conference on Learning Representations (**ICLR**), 2025. [Oral · Top 1.8%](#)

[C2] Stochastic Optimal Control for Diffusion Bridges in Function Spaces.

Byoungwoo Park, Jungwon Choi, Sungbin Lim†, Juho Lee†
Advances in Neural Information Processing Systems (**NeurIPS**), 2024.

[C1] Neural Stochastic Differential Games for Time-series Analysis.

Sungwoo Park*, **Byoungwoo Park***, Moontae Lee, Changhee Lee
International Conference on Machine Learning (**ICML**), 2023.

Journal Papers

[J1] Multi-Marginal Schrödinger Bridge Matching.
Byoungwoo Park and Juho Lee
Transactions on Machine Learning Research (TMLR), 2026.

Workshops Papers & Preprints

[P3] LongTake: Learning to Sustain Dynamics in Long-Horizon Video Generation
Byoungwoo Park, Jaemoo Choi, Juho Lee[†], Yongxin Chen[†]
Submitted, 2026

[P2] Proper Scoring Rule-based Diffusion for Probabilistic Weather Forecasting
Joonhyeong Park, Giung Nam, Hyungi Lee, Kyunghyun Cho, **Byoungwoo Park**[†], Juho Lee[†]
Submitted, 2026

[P1] Diffusion Policy Improvement with Proposal-Conditioned Refinement Flows
Junhyun Ha, Juho Lee, **Byoungwoo Park**
Submitted, 2026

[W1] Improving Constrained Language Generation via Self-Distilled Twisted Sequential Monte Carlo.
Sooyeon Kim, Giung Nam, **Byoungwoo Park**, Juho Lee
NeurIPS 2025 Workshop on Frontiers in Probabilistic Inference: Learning meets Sampling.

TALKS

Amortized Control of Continuous State Space Feynman-Kac Model for Irregular Time Series.

Oral presentation at ICLR

04/2025

Global AI Frontier Lab

02/2025

AWARDS

Excellence Scholarship for Ph.D. Students in STEM,
Ministry of Science and ICT (MSIT) and Korea Student Aid Foundation (KOSAF), 2026.

NeurIPS Scholar Award,
Neural Information Processing Systems (NeurIPS), 2024.

ACADEMIC SERVICE

Reviewer: ICML [2022,2026-], ICLR [2025-], NeurIPS [2025-], TMLR [2025]